

Quantum Finance



ANANTWAVE

Internship Report

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In Cooperation with

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Abstract

This report investigates the application of **quantum computing** to real-world **financial problems**, with a focus on **hybrid quantum-classical algorithms** implemented using **PennyLane**. Key areas explored include **portfolio optimization** using the **Variational Quantum Eigensolver (VQE)**, graph partitioning with the **Quantum Approximate Optimization Algorithm (QAOA)** for solving the **MaxCut problem**, and **financial modeling** using the **Quantum Boltzmann Machine (QBM)**.

The work also includes experimental studies on foundational algorithms such as the **Deutsch–Jozsa** and **Shor’s algorithm** to understand their relevance to **quantum advantage**. Practical implementations were carried out in both **Google Colab** and **Visual Studio Code**, leveraging **PennyLane’s simulators** for efficient quantum circuit testing. This project demonstrates how **quantum algorithms** can be effectively integrated with **classical methods** to address complex financial tasks like **risk-aware asset selection** and **combinatorial optimization**, establishing a foundation for future **quantum-enhanced financial applications**.

Contents

1	Introduction	5
2	State of Art	6
3	Methodology	7
3.1	Frameworks and Platforms Used	7
3.2	Approach to Algorithms and Protocols	7
4	Implementation of Algorithms	8
4.1	Exploring the MaxCut Problem Using QAOA and PennyLane	9
4.1.1	Overview:	9
4.1.2	Algorithm:	9
4.1.3	Code and Execution	10
4.2	Quantum Boltzmann Machine (QBM)	12
4.2.1	Overview:	12
4.2.2	Algorithm:	12
4.2.3	Code and Execution:	13
4.2.4	Technical Analysis:	15
4.3	Variational Quantum Eigensolver (VQE)	16
4.3.1	Overview:	16
4.3.2	VQE Algorithm in Quantum Finance:	17
4.3.3	Code and Execution	18
4.3.4	Output	19
4.3.5	Technical Analysis	22
4.4	Deutsch-Jozsa Algorithm	24
4.4.1	Problem Definition and Theoretical Background	24
4.4.2	Quantum Circuit Implementation	24
4.4.3	Quantum Monte Carlo Applications	24
4.4.4	Detailed Option Pricing Example	25
4.4.5	Complexity Analysis	25
4.4.6	Implementation Considerations	25
4.4.7	Code and Execution:	25
4.5	Shor's Algorithm for Integer Factorization	26
4.5.1	Overview	26
4.5.2	Algorithm Implementation	26
4.5.3	Complexity Comparison	26
4.5.4	Code and Execution:	27
5	Implementation	28
5.1	System Design	28
5.2	Coding and Simulation	28

5.3	Execution Process	28
5.4	Optimization Steps	29
5.5	Tools and Frameworks	29
5.6	Challenges and Mitigation	29
6	Conclusion	30
7	Bibliography	31

List of Figures

1	MaxCut QAOA implementation results showing optimal graph partition	11
2	Flowchart of MaxCut QAOA Algorithm	12
3	QBM training metrics showing convergence and parameter optimization .	14
4	Final QBM results demonstrating quantum advantage in financial modeling	15
5	Flowchart of Quantum Boltzmann Machine (QBM) Training Algorithm .	16
6	Bar graph of optimal portfolio weights for assets TCS.NS, SIEMENS.NS, NHPC.NS, and IDEA.NS	20
7	Tabular display of 16 fundamental parameters for portfolio assets	20
8	Result of POST Request(quantum/optimize)	20
9	Result of POST Request(quantum/optimize)	21
10	Frontend representation of the app displaying parameters	21
11	Flowchart of Variational Quantum Eigensolver (VQE) Algorithm in Quantum Finance	22
12	Quantum Monte Carlo simulation results for option pricing probability estimation	26
13	Periodicity detection for $18^z \bmod 77$ showing the repeating pattern every 30 units, which is crucial for factorization in Shor's algorithm	27
14	Successful factorization results showing the prime factors 11 and 7 of $N = 77$ found using period $r = 30$ and intermediate value $x = 43$	27

1 Introduction

Quantum computing holds immense potential for transforming the financial industry by solving complex problems at a speed and scale unattainable by classical computers. The financial sector is characterized by vast datasets, intricate optimization challenges, and the need for rapid decision-making, making it an ideal domain for quantum technologies.

This report explores quantum finance, which leverages quantum algorithms to enhance various financial applications such as portfolio optimization, risk management, algorithmic trading, and derivatives pricing. The significance of this field lies in its potential to revolutionize financial modeling, security, and computational efficiency. Traditional financial models often face limitations in processing power and accuracy, whereas quantum-enhanced techniques promise breakthroughs in predictive analytics, fraud detection, and market simulation.

The objective of this study is to provide a comprehensive understanding of quantum finance by covering its theoretical foundations, real-world applications, and implementation methodologies using tools like PennyLane. Additionally, the report discusses the broader landscape of quantum technologies, including quantum cryptography for secure transactions, quantum machine learning for advanced financial insights, and the impact of quantum artificial intelligence (AI) in financial decision-making.

By outlining these advancements, this report aims to establish the relevance and transformative potential of quantum computing in modern finance, setting the stage for further research and practical adoption of quantum-enhanced financial strategies.

2 State of Art

Quantum computing is transforming financial problem-solving by enhancing optimization, risk assessment, and pricing models. Classical financial algorithms struggle with the complexity and computational demands of large-scale financial systems, while quantum algorithms offer significant speed and efficiency advantages.

One of the most promising approaches in quantum finance is the **Quantum Approximate Optimization Algorithm (QAOA)**, which efficiently tackles portfolio optimization problems by balancing risk and return. Similarly, the **Variational Quantum Eigensolver (VQE)** is applied to optimize asset allocation and solve complex financial equations. These quantum methods outperform classical techniques by exploring multiple possibilities simultaneously, reducing computational overhead.

In **risk management**, quantum Monte Carlo methods accelerate simulations used for calculating risk measures like **Value at Risk (VaR)** and **Conditional Value at Risk (CVaR)**. Quantum speedup in these calculations enables more accurate and real-time risk assessments, which are crucial in volatile financial markets.

For **derivatives pricing**, quantum algorithms improve models like the **Black-Scholes equation** by using quantum parallelism to evaluate multiple price scenarios faster than classical Monte Carlo simulations. Additionally, **quantum path integral methods** are being explored for simulating asset price movements with greater precision.

Current quantum finance research is supported by platforms like **IBM Qiskit**, **PennyLane**, and **D-Wave Leap**, which provide tools for developing and testing quantum financial models. Hybrid quantum-classical systems are also gaining traction, leveraging classical computing power for preprocessing and quantum computing for speedups in complex financial tasks.

Despite these advancements, challenges remain. **Quantum hardware limitations**, such as **qubit decoherence**, **error correction**, and **scalability**, restrict the practical deployment of quantum financial models. Most applications are still in the experimental phase, requiring hybrid approaches until fault-tolerant quantum systems become viable.

The progress in quantum finance highlights its potential to revolutionize financial decision-making, risk management, and trading strategies, paving the way for more efficient and intelligent financial models in the future.

3 Methodology

This section details the approach used to explore quantum finance applications, including the platforms, algorithms, and implementation strategies employed in this study.

3.1 Frameworks and Platforms Used

The implementation was carried out using **Google Colab**, utilizing **PennyLane**, a quantum machine learning framework that enables hybrid quantum-classical computing. The quantum simulations were performed on the default.qubit simulator provided by PennyLane, allowing efficient prototyping of quantum circuits without requiring access to actual quantum hardware. Additionally, **NumPy** was used for classical computations and data handling. For the Variational Quantum Eigensolver (VQE) implementation, **Visual Studio Code (VS Code)** was used as the primary development environment, providing enhanced debugging capabilities and integration with quantum computing libraries.

3.2 Approach to Algorithms and Protocols

Several quantum algorithms were explored for financial modeling:

- **Quantum Approximate Optimization Algorithm (QAOA):** Used for portfolio optimization, where it efficiently selects an asset allocation strategy by solving combinatorial optimization problems.
- **Quantum Boltzmann Machine (QBM):** A quantum generalization of classical Boltzmann machines that leverages quantum effects for enhanced sampling and learning in financial data analysis.
- **Variational Quantum Eigensolver (VQE):** Used to solve portfolio optimization as an eigenvalue problem, VQE approximates the ground state of a Hamiltonian representing the financial system.
- **Deutsch-Jozsa Algorithm:** Explored for quantum advantage demonstration.
- **Shor's Algorithm:** Explored for factorization applications.

Each algorithm was selected based on its ability to outperform classical methods in speed or accuracy, particularly for solving complex financial problems.

4 Implementation of Algorithms

This section provides comprehensive technical details of all quantum algorithms implemented, along with their execution methodologies and practical considerations.

4.1 Exploring the MaxCut Problem Using QAOA and PennyLane

In this project, I implemented the **Quantum Approximate Optimization Algorithm (QAOA)** to solve the **MaxCut problem** using a quantum computing framework called **PennyLane**.

4.1.1 Overview:

The MaxCut problem involves dividing a graph into two groups such that the number of edges between the groups is maximized. To solve this using QAOA, I encoded the MaxCut problem into a quantum Hamiltonian (a mathematical operator that represents the energy of the system). The QAOA algorithm works by alternating between two quantum operations: the **cost unitary**, which encodes the MaxCut problem, and the **mixer unitary**, which explores different possible solutions. These operations are applied multiple times, with parameters (gamma and beta) optimized to maximize the number of edges in the cut. Finally, the quantum circuit is measured to obtain a bitstring representing the solution, and through a classical optimizer, the best parameters are found to approximate the optimal cut. This implementation was run on a simulator in Google Colab, making use of PennyLane's quantum circuit capabilities.

4.1.2 Algorithm:

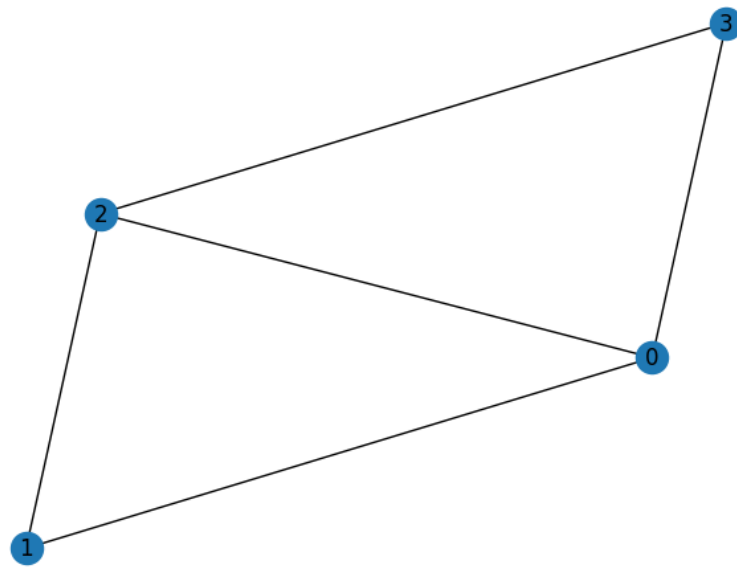
- **Goal:** Divide a graph into two sets to maximize the number of edges connecting the sets (MaxCut), using quantum optimization.
- **Define the Graph:**
 - Choose or create a graph $G = (V, E)$ with nodes and edges.
 - This graph represents the problem input.
- **Create the Cost Hamiltonian:**
 - Convert the MaxCut objective into a Hamiltonian using PauliZ operators.
 - For each edge (i, j) , encode its contribution to the MaxCut.
- **Initialize Quantum Device:**
 - Set up a quantum simulator (e.g., `default.qubit`) with the number of qubits equal to the number of graph nodes.
 - Use `shots=1` for sampling a single outcome per run.
- **Build the QAOA Circuit:**
 - Apply Hadamard gates to all qubits to create superposition.
 - Apply cost unitary based on the graph structure using CNOT and RZ.
 - Apply mixer unitary using RX gates.
 - Repeat the above unitaries for p layers using trainable parameters γ (gamma) and β (beta).
- **Optimize Parameters:**

- Use a classical optimizer (e.g., gradient descent) to minimize the expectation value of the Hamiltonian.
- Iterate to find the best parameters that maximize the cut.
- **Sample the Final Circuit:**
 - Use the trained circuit to sample bitstrings.
 - Each bitstring represents a possible solution (i.e., a way to divide the graph nodes into two sets).
- **Find the Best Cut:**
 - Count how many times each bitstring appears in the sampled results.
 - The most frequent bitstring is likely the best or near-best solution to the MaxCut problem.

4.1.3 Code and Execution

The implementation details and results are as follows:

- **Repository:** <https://github.com/AnantWave/quantum-finance-lab-Khushboo-project.git>
- **Clone Command:**
`git clone https://github.com/AnantWave/quantum-finance-lab-Khushboo-project.git`
- **Execution:** Navigate to `MAXCUT_QAOA.ipynb` and run the notebook
- **Output:** The experimental results are shown in Figure 1 (see page 11)



```
Step 0: Cost = -3.0000  
Step 10: Cost = -2.0000  
Step 20: Cost = 0.0000  
Step 30: Cost = -4.0000  
Step 40: Cost = 0.0000  
Step 50: Cost = -4.0000  
Step 60: Cost = -4.0000  
Step 70: Cost = -3.0000  
Step 80: Cost = -4.0000  
Step 90: Cost = -4.0000  
Most frequent bitstring (cut): 1110
```

Figure 1: MaxCut QAOA implementation results showing optimal graph partition

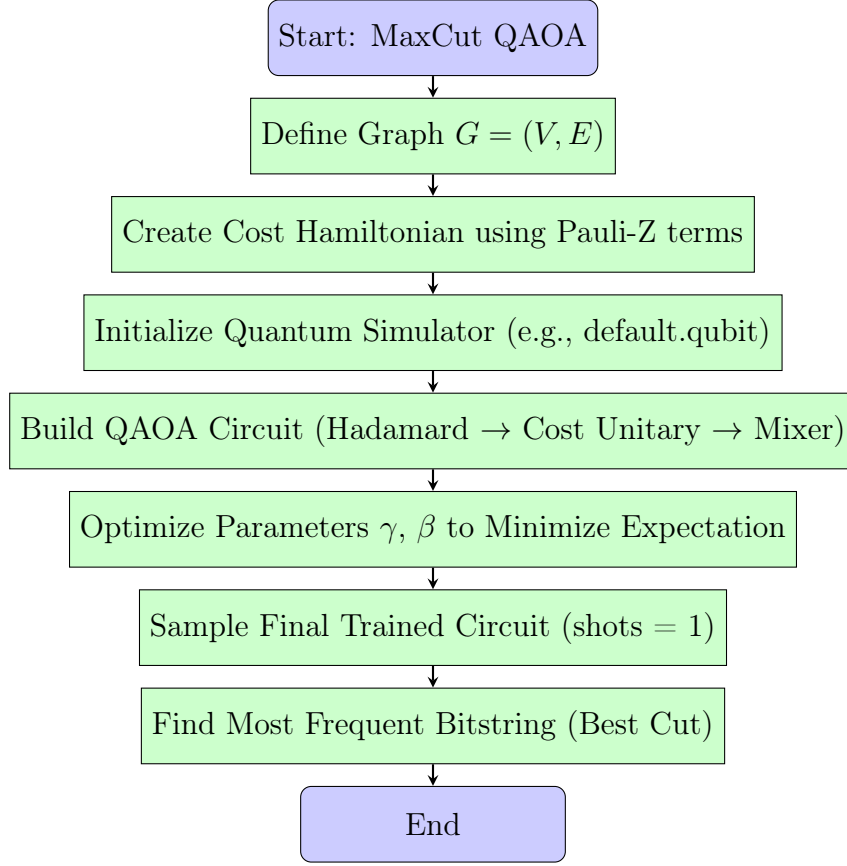


Figure 2: Flowchart of MaxCut QAOA Algorithm

4.2 Quantum Boltzmann Machine (QBM)

4.2.1 Overview:

The Quantum Boltzmann Machine (QBM) is a quantum analog of classical Boltzmann Machines, leveraging qubits and quantum mechanical principles like superposition and entanglement for enhanced machine learning capabilities. QBMs model probability distributions by constructing a Hamiltonian system of interacting qubits, offering potential advantages in processing complex, high-dimensional financial data.

4.2.2 Algorithm:

1. Initialize the QBM Model:

- (a) Set the number of qubits in the QBM (e.g., 3 qubits)
- (b) Initialize quantum circuit parameters (rotation angles)
- (c) Define Hamiltonian using quantum operators (Pauli Z/X) with coefficients

2. **Quantum State Preparation:**
 - (a) Apply Hadamard gates for superposition
 - (b) Apply RX rotations based on model parameters
3. **Define the Cost Function:**
 - (a) Compute expectation value of Hamiltonian
4. **Gibbs Sampling:**
 - (a) Initialize quantum state from data
 - (b) Perform iterative state updates
5. **Reconstruct the State:**
 - (a) Apply additional rotations (RX/RZ)
6. **Calculate Energies:**
 - (a) Compute original and reconstructed state energies
7. **Energy Difference:**
 - (a) Calculate gradient for parameter updates
8. **Update Parameters:**
 - (a) Apply Gradient Descent optimization
9. **Training Loop:**
 - (a) Repeat for multiple epochs
 - (b) Monitor coefficient updates
10. **Finalize Model:**
 - (a) Deploy trained QBM for inference tasks

4.2.3 Code and Execution:

- **Repository:** <https://github.com/AnantWave/quantum-finance-lab-Khushboo-project.git>
- **Clone Command:** `git clone https://github.com/AnantWave/quantum-finance-lab-Khushboo-project.git`
- **Execution:** Navigate to `QBM.ipynb` and run the notebook.
- **Output:** The experimental results are shown in Figure 3 (training metrics) and Figure 4 (final results) on pages 14 and 15 respectively

```

Hamiltonian: 0.5 * Z(0) + -1.2 * Z(1)
State: [0.33776245-0.10448217j 0.33776245-0.10448217j 0.33776245-0.10448217j
0.33776245-0.10448217j 0.33776245-0.10448217j 0.33776245-0.10448217j
0.33776245-0.10448217j 0.33776245-0.10448217j]
Epoch 0: Optimized coefficients: [0.5, -1.2]
Epoch 1: Optimized coefficients: [0.5, -1.2]
Epoch 2: Optimized coefficients: [0.5, -1.2]
Epoch 3: Optimized coefficients: [0.5, -1.2]
Epoch 4: Optimized coefficients: [0.5, -1.2]
Epoch 5: Optimized coefficients: [0.5, -1.2]
Epoch 6: Optimized coefficients: [0.5, -1.2]
Epoch 7: Optimized coefficients: [0.5, -1.2]
Epoch 8: Optimized coefficients: [0.5, -1.2]
Epoch 9: Optimized coefficients: [0.5, -1.2]
Epoch 10: Optimized coefficients: [0.5, -1.2]
Epoch 11: Optimized coefficients: [0.5, -1.2]
Epoch 12: Optimized coefficients: [0.5, -1.2]
Epoch 13: Optimized coefficients: [0.5, -1.2]
Epoch 14: Optimized coefficients: [0.5, -1.2]
Epoch 15: Optimized coefficients: [0.5, -1.2]
Epoch 16: Optimized coefficients: [0.5, -1.2]
Epoch 17: Optimized coefficients: [0.5, -1.2]
Epoch 18: Optimized coefficients: [0.5, -1.2]
Epoch 19: Optimized coefficients: [0.5, -1.2]
Epoch 20: Optimized coefficients: [0.5, -1.2]
Epoch 21: Optimized coefficients: [0.5, -1.2]
Epoch 22: Optimized coefficients: [0.5, -1.2]
Epoch 23: Optimized coefficients: [0.5, -1.2]
Epoch 24: Optimized coefficients: [0.5, -1.2]
Epoch 25: Optimized coefficients: [0.5, -1.2]
Epoch 26: Optimized coefficients: [0.5, -1.2]
Epoch 27: Optimized coefficients: [0.5, -1.2]
Epoch 28: Optimized coefficients: [0.5, -1.2]
Epoch 29: Optimized coefficients: [0.5, -1.2]
Epoch 30: Optimized coefficients: [0.5, -1.2]
Epoch 31: Optimized coefficients: [0.5, -1.2]
Epoch 32: Optimized coefficients: [0.5, -1.2]
Epoch 33: Optimized coefficients: [0.5, -1.2]
Epoch 34: Optimized coefficients: [0.5, -1.2]
Epoch 35: Optimized coefficients: [0.5, -1.2]
Epoch 36: Optimized coefficients: [0.5, -1.2]
Epoch 37: Optimized coefficients: [0.5, -1.2]
Epoch 38: Optimized coefficients: [0.5, -1.2]
Epoch 39: Optimized coefficients: [0.5, -1.2]
Epoch 40: Optimized coefficients: [0.5, -1.2]
Epoch 41: Optimized coefficients: [0.5, -1.2]
Epoch 42: Optimized coefficients: [0.5, -1.2]
Epoch 43: Optimized coefficients: [0.5, -1.2]
Epoch 44: Optimized coefficients: [0.5, -1.2]
Epoch 45: Optimized coefficients: [0.5, -1.2]
Epoch 46: Optimized coefficients: [0.5, -1.2]
Epoch 47: Optimized coefficients: [0.5, -1.2]
Epoch 48: Optimized coefficients: [0.5, -1.2]
Epoch 49: Optimized coefficients: [0.5, -1.2]

```

Figure 3: QBM training metrics showing convergence and parameter optimization

```

Epoch 93: Optimized coefficients: [0.5, -1.2]
Epoch 94: Optimized coefficients: [0.5, -1.2]
Epoch 95: Optimized coefficients: [0.5, -1.2]
Epoch 96: Optimized coefficients: [0.5, -1.2]
Epoch 97: Optimized coefficients: [0.5, -1.2]
Epoch 98: Optimized coefficients: [0.5, -1.2]
Epoch 99: Optimized coefficients: [0.5, -1.2]
Samples: [[1 0 1]
[0 1 1]
[0 1 1]
...
[0 1 0]
[1 1 0]
[0 1 1]]
State with data: [0.+0.j 0.+0.j 0.+0.j 0.+0.j 0.+0.j 1.+0.j 0.+0.j 0.+0.j]
Expectation value: 0.9950041652780256

```

Figure 4: Final QBM results demonstrating quantum advantage in financial modeling

4.2.4 Technical Analysis:

Advantages:

- Quantum parallelism for faster processing
- Efficient high-dimensional representation
- Enhanced state space exploration
- Direct energy minimization

Limitations:

- Current hardware constraints (noise, qubit count)
- Steep learning curve
- Algorithm immaturity

Comparison:

- Classical: More stable and scalable
- Quantum: Better for complex data patterns

Usage Guidelines:

- Recommended for quantum advantage research
- Not yet production-ready for finance

Future Directions:

- Hardware improvements
- Algorithm refinement
- Hybrid classical-quantum approaches

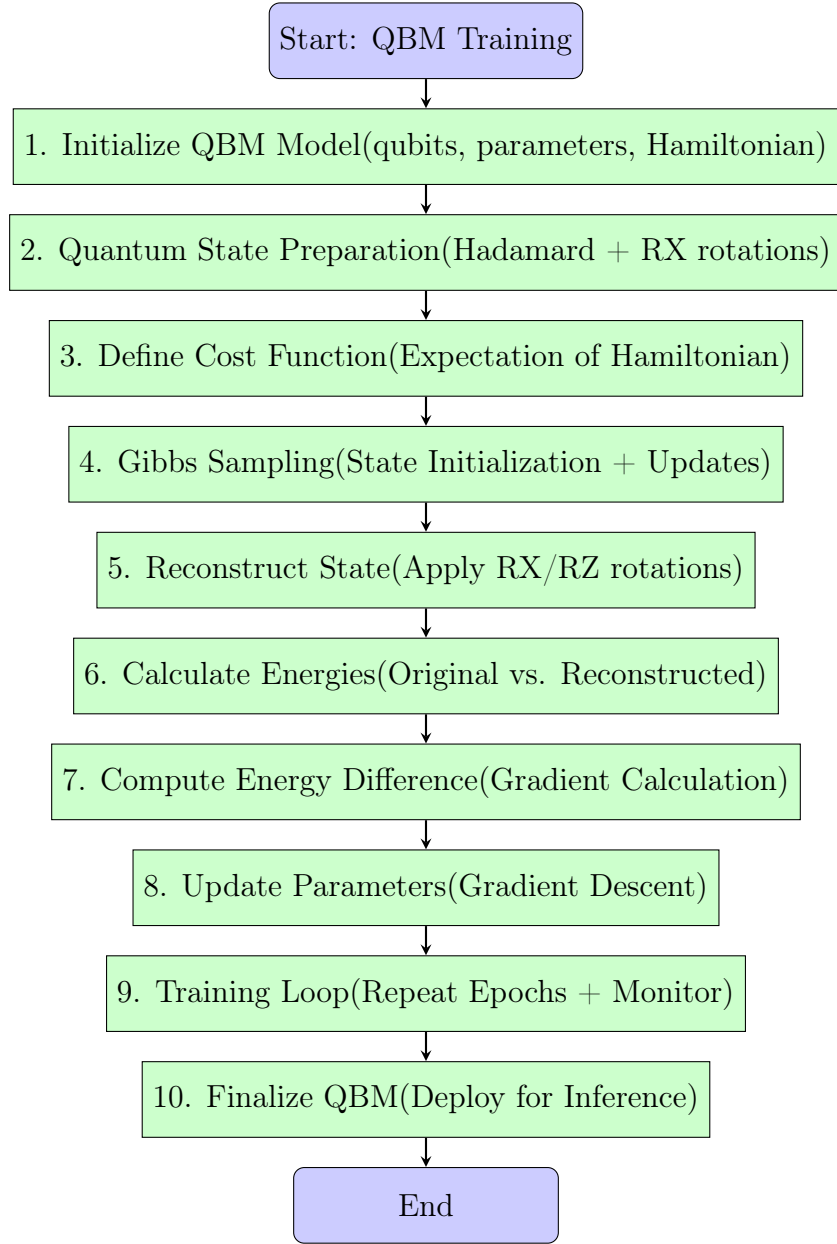


Figure 5: Flowchart of Quantum Boltzmann Machine (QBM) Training Algorithm

4.3 Variational Quantum Eigensolver (VQE)

4.3.1 Overview:

The **Variational Quantum Eigensolver (VQE)** is a quantum algorithm used primarily for finding the **lowest energy state (ground state)** of a quantum system, which is described by its **Hamiltonian**. It is especially useful for quantum chemistry and material science simulations, where the goal is to find the properties of molecules or materials by

solving their quantum mechanical Hamiltonians.

In the field of **quantum finance**, the **Variational Quantum Eigensolver (VQE)** can be a powerful tool for solving optimization problems, pricing derivatives, or finding the ground state of complex financial models. Quantum finance aims to leverage quantum computing techniques to tackle finance-related problems that are difficult or impossible for classical computers to handle efficiently.

VQE can be particularly useful for problems such as **option pricing**, **portfolio optimization**, **risk analysis**, and **large-scale Monte Carlo simulations**. Here's how VQE can be applied in quantum finance:

In quantum finance, we are typically interested in **finding the optimal parameters** for financial models. These models can involve complex interactions, large datasets, and intricate computations that are resource-heavy for classical machines.

The code implementation shows how Qiskit, IBM's quantum computing framework, was utilized to implement the VQE algorithm for portfolio optimization.

4.3.2 VQE Algorithm in Quantum Finance:

1. Initialization of the Ansatz (Quantum Circuit):

- Choose a parameterized quantum circuit (ansatz) that encodes the classical financial data (such as option strike price, interest rate, and volatility) into quantum states.
- This quantum circuit might include rotations and entangling gates that depend on financial parameters. These parameters are adjusted during the optimization.

2. Define the Hamiltonian:

- The **Hamiltonian** represents the energy of the financial model. In finance, it could represent the cost of holding an option or the price of an asset.

3. Measurement (Expectation Value of the Hamiltonian):

- After applying the quantum circuit, we measure the quantum state to compute the expectation value of the Hamiltonian. This is the quantum analog of calculating the energy of the system.
- In quantum finance, this **expectation value** corresponds to the **price** or **value** of the financial instrument we're modeling. For example, it could be the option price or the expected return of a portfolio.

4. Classical Optimization:

- The **classical optimizer** adjusts the parameters of the quantum circuit to minimize the expectation value of the Hamiltonian (the financial model's energy or price).

- The optimizer iterates, adjusting the parameters of the quantum circuit, such as the portfolio weights or volatility estimates, to find the optimal solution that minimizes the error in the model.

5. Convergence:

- The optimization process continues until the energy (price) converges to a minimum value, which represents the optimal solution for the quantum financial model.
- Once convergence is achieved, the optimized parameters (e.g., optimal portfolio weights or option price) are extracted from the quantum circuit.

4.3.3 Code and Execution

- **Repository:** <https://github.com/ManthriParthiv/fastapi-stock-app>
- **Clone Command:**

```
git clone https://github.com/ManthriParthiv/fastapi-stock-app.git
cd fastapi-stock-app
```

Execution Sequence:

1. Install Dependencies:

```
pip install -r fastapi-stock-app/Backend-Server/
quantum_optimizer/requirements.txt
```

2. Data Preparation:

```
cd fastapi-stock-app/Backend-Server/quantum_optimizer

python -m preprocessing.fetch_data.py
python -m preprocessing.fetch_fundamentals.py
python -m preprocessing.parallel_fetch.py
```

3. Quantum Optimization:

```
python -m processing.vqe_portfolio.py
python run_all.py
```

4. Analysis and Visualization:

```
python -m postprocessing.analyze.py
python -m postprocessing.visualize.py
```

5. Start Backend Server:

```
cd ../../Backend_Server
uvicorn main:app --reload
```

6. Start Frontend UI (New Terminal):

```
cd ../frontend_UI/Stocks_App
npm install
npm run dev
```

7. Access Application:

- Backend API: <http://localhost:8000>
- Frontend UI: <http://localhost:3000>
- API Documentation: <http://localhost:8000/docs>

Architecture Flow:

1. Data fetching scripts populate **data/** directory with market data
2. Quantum optimization processes generate portfolio weights
3. FastAPI backend serves optimization results via REST API
4. React frontend provides interactive user interface
5. Real-time optimization available through web interface

Key Endpoints:

- **POST /optimize** - Portfolio optimization with quantum VQE
- **GET /results** - Retrieve optimization results
- **GET /fundamentals** - Access financial parameter data

Notes: All other details are present in **NOTES.txt**.

4.3.4 Output

Parameters Fundamental parameters analyzed (16 total):

- Part 1: Weight, P/E, P/B, RDE, Curr, D/E, EV/EBITDA, Beta, MktCap
- Part 2: RevGrowth, PEG, NetMargin, FreeCF, OpMargin, P/S, Payout, CurrRatio

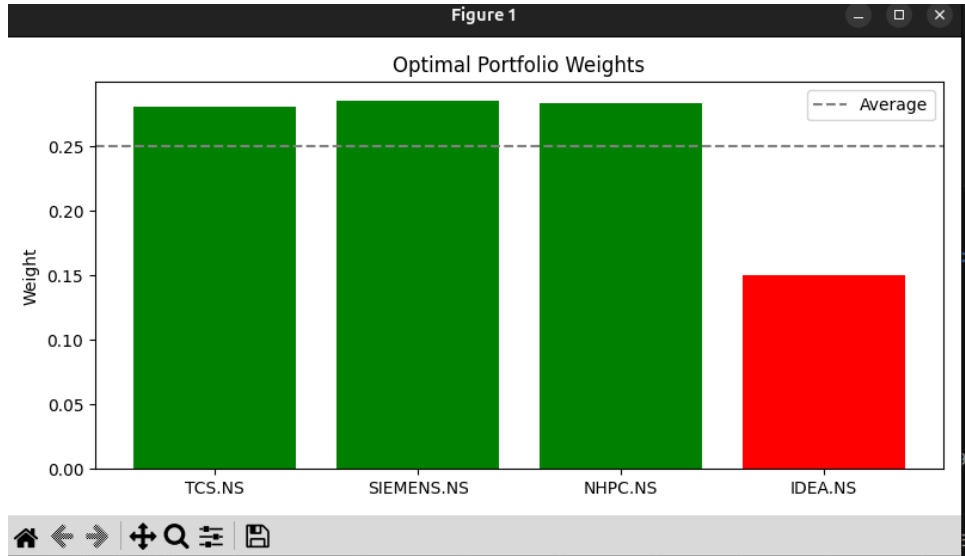


Figure 6: Bar graph of optimal portfolio weights for assets TCS.NS, SIEMENS.NS, NHPC.NS, and IDEA.NS

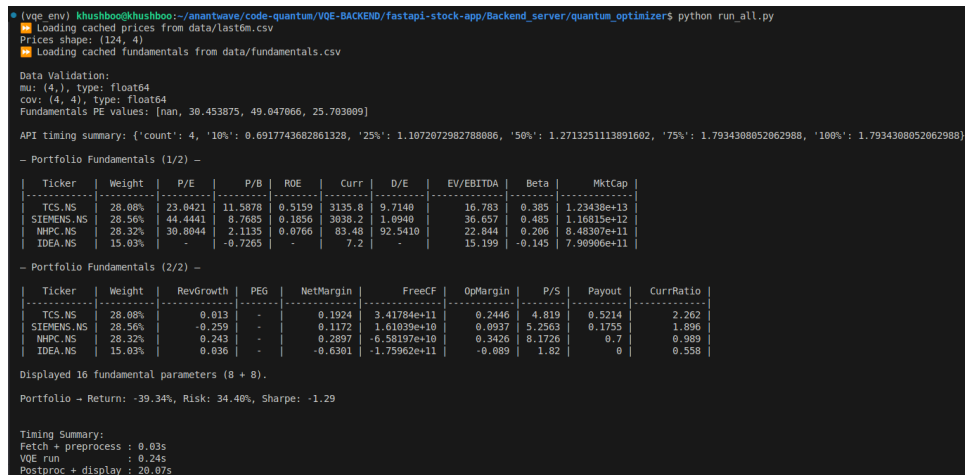


Figure 7: Tabular display of 16 fundamental parameters for portfolio assets

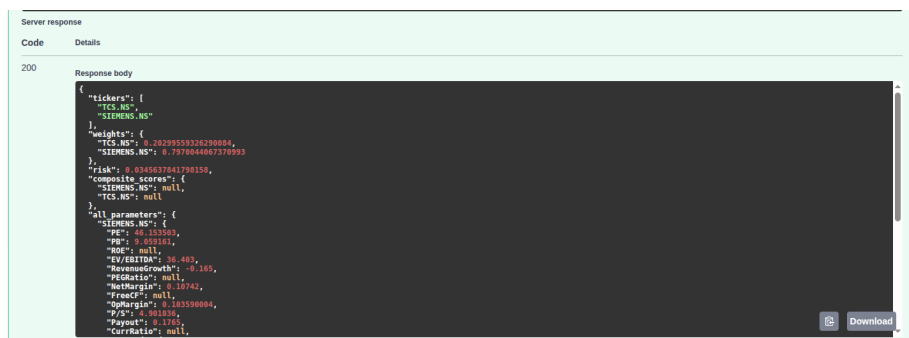


Figure 8: Result of POST Request(quantum/optimize)



Figure 9: Result of POST Request(quantum/optimize)

Ticker	Recommended Weight
SIEMENS.NS	0.2024280407616788%
TCL.NS	0.797571959238321%
Overall Risk: 0.013825792271920241%	
All Parameters	
SIEMENS.NS	
Parameter	Value
PE	46.153503
PB	9.029161
ROE	
EV/EBITDA	36.403
RevenueGrowth	-0.165
PEGRatio	
NetMargin	0.10742
FreeCF	
OpMargin	0.103590004
P/S	4.901036
Payoff	0.1765
CurrRatio	
LastUpdated	2025-09-01 19:57:50
TCL.NS	
Parameter	Value
PE	22.880037
PB	11.502034
ROE	0.51589
EV/EBITDA	16.502
RevenueGrowth	0.013
PEGRatio	
NetMargin	0.19236
FreeCF	341783740416
OpMargin	0.24457
P/S	4.396547
Payoff	0.3214
CurrRatio	2.262
LastUpdated	2025-09-01 19:57:51

Figure 10: Frontend representation of the app displaying parameters

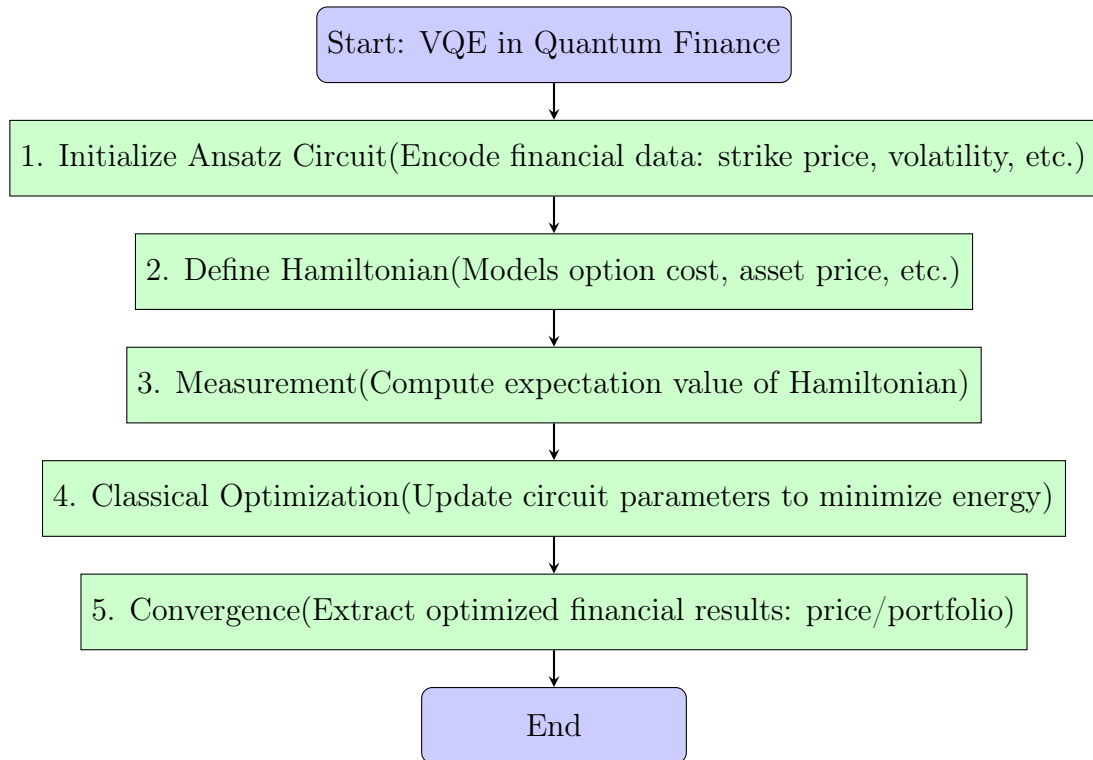


Figure 11: Flowchart of Variational Quantum Eigensolver (VQE) Algorithm in Quantum Finance

4.3.5 Technical Analysis

Financial Applications:

1. Portfolio optimization
2. Option pricing
3. Risk metric calculation

Performance Metrics:

1. Convergence iterations
2. Computational speed
3. Solution accuracy

Advantages:

1. Handles non-convex optimization landscapes
2. Scalable to intermediate problem sizes

3. Noise-resilient through error mitigation

Limitations:

1. Requires careful ansatz design
2. Sensitive to optimizer choice
3. Limited by current quantum hardware

4.4 Deutsch-Jozsa Algorithm

4.4.1 Problem Definition and Theoretical Background

The Deutsch-Jozsa algorithm is a fundamental quantum algorithm that solves a specific problem designed to showcase quantum advantage. The problem is formulated as follows:

Consider a Boolean function:

$$f : \{0, 1\}^n \rightarrow \{0, 1\}$$

that satisfies exactly one of these properties:

- **Constant:** All inputs map to the same output ($f(x) = 0$ for all x or $f(x) = 1$ for all x)
- **Balanced:** Exactly half of the inputs map to 0 and half map to 1

The key insight is that while a classical computer must check more than half of all possible inputs in the worst case, a quantum computer can determine the nature of the function with certainty using just one evaluation through quantum parallelism.

4.4.2 Quantum Circuit Implementation

The algorithm works by:

1. Initializing $n + 1$ qubits: n input qubits in state 0 and 1 output qubit in state 1
2. Applying Hadamard gates to all qubits, creating superposition:

$$\frac{1}{\sqrt{2^n}} \sum_{x \in \{0,1\}^n} x \otimes \frac{0 - 1}{\sqrt{2}}$$

3. Evaluating the oracle U_f which computes $f(x)$ and stores it in phase:

$$\frac{1}{\sqrt{2^n}} \sum_x (-1)^{f(x)} x \otimes \frac{0 - 1}{\sqrt{2}}$$

4. Applying Hadamard gates again and measuring

The measurement will yield $0^{\otimes n}$ if f is constant, and any other state if balanced.

4.4.3 Quantum Monte Carlo Applications

Mathematical Foundations of Amplitude Estimation The quantum Monte Carlo approach builds on amplitude estimation, which combines:

- **Quantum Phase Estimation:** For determining eigenvalues of unitary operators
- **Grover's Algorithm:** For amplitude amplification

The core mathematical operation can be expressed as:

$$Q = -AS_0A^{-1}S_\chi$$

where:

- A is the state preparation operator
- S_0 and S_χ are reflection operators
- Each application of Q performs one Grover iteration

4.4.4 Detailed Option Pricing Example

For our binary option model:

- Underlying asset price S_t follows binomial model
- Risk-neutral probability p calculated via:

$$p = \frac{e^{r\Delta t} - d}{u - d}$$

where u, d are up/down factors, r is risk-free rate

- Quantum state preparation encodes this probability:

$$\psi = \sqrt{1-p}0 + \sqrt{p}1$$

- The payoff operator $\mathcal{P}$ acts as:

$$\mathcal{P}x0 = x\text{payoff}(x)$$

4.4.5 Complexity Analysis

The quadratic speedup comes from:

- Classical Monte Carlo error $\epsilon \sim O(1/\sqrt{M})$ for M samples
- Quantum error $\epsilon \sim O(1/M)$ using M applications of Q
- For target error ϵ , classical needs $O(1/\epsilon^2)$ samples vs quantum $O(1/\epsilon)$

4.4.6 Implementation Considerations

Key practical aspects include:

- State preparation circuits must be efficient
- Payoff functions need quantum arithmetic implementation
- Error mitigation techniques are crucial for noisy hardware
- Quantum advantage appears for high-dimensional problems

4.4.7 Code and Execution:

- **Repository:** <https://github.com/AnantWave/quantum-finance-lab-Khushboo-project.git>
- **Clone Command:** `git clone https://github.com/AnantWave/quantum-finance-lab-Khushboo-project.git`
- **Execution:** Navigate to `First_round_of_quantum_Monte_Carlo_oneperiod_option_payoff_using_DEUTSCH_JOZSA.ipynb` and run the notebook.
- **Output:** The experimental results are shown in Figure 12 on page 26.

```
Estimated probability = 0.0000 (true p=0.3)
95% Confidence interval: (0.0, 0.0)
```

Figure 12: Quantum Monte Carlo simulation results for option pricing probability estimation

4.5 Shor's Algorithm for Integer Factorization

4.5.1 Overview

Shor's algorithm demonstrates quantum advantage for integer factorization, a problem central to modern cryptography. The implementation focuses on factoring small composite numbers using quantum circuits.

4.5.2 Algorithm Implementation

Mathematical Steps

1. Choose random a where $1 < a < N$ and $\gcd(a, N) = 1$
2. If $\gcd(a, N) > 1$, return as factor
3. Find period r of $a^x \bmod N$ using:
 - Quantum Fourier Transform (QFT)
 - Modular exponentiation
4. Verify r satisfies:
 - r must be even
 - $a^{r/2} \not\equiv -1 \pmod{N}$
5. Compute factors: $\gcd(a^{r/2} \pm 1, N)$

4.5.3 Complexity Comparison

- **Classical Methods:**
 - Best known: General Number Field Sieve
 - Complexity: $\exp(O((\log N)^{1/3}(\log \log N)^{2/3}))$
- **Shor's Algorithm:**
 - Time: $O((\log N)^3)$
 - Space: $O(\log N)$ qubits

4.5.4 Code and Execution:

- **Repository:** <https://github.com/AnantWave/quantum-finance-lab-Khushboo-project.git>
- **Clone Command:** `git clone https://github.com/AnantWave/quantum-finance-lab-Khushboo-project.git`
- **Execution:** Navigate to `Shor's_algorithm.ipynb` and run the notebook.
- **Output:** The experimental results are shown in Figure 13 (period finding) and Figure 14 (factorization results) on pages 27 and 27 respectively.

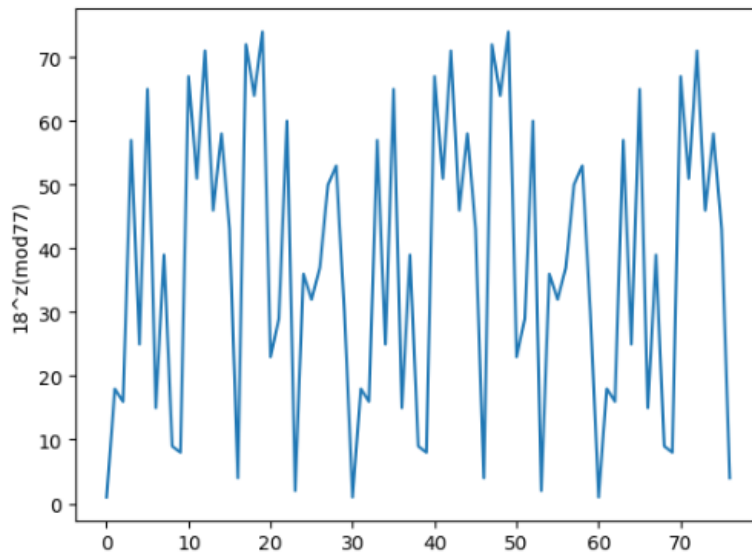


Figure 13: Periodicity detection for $18^z \pmod{77}$ showing the repeating pattern every 30 units, which is crucial for factorization in Shor's algorithm

$$x=43.0$$
$$11 \quad 7$$

Figure 14: Successful factorization results showing the prime factors 11 and 7 of $N = 77$ found using period $r = 30$ and intermediate value $x = 43$

5 Implementation

The implementation section outlines the hands-on execution of quantum finance techniques using both Qiskit and PennyLane frameworks. It includes system architecture, coding methodology, simulation environments, and optimization procedures employed to solve financial and combinatorial problems using variational quantum algorithms like VQE, QAOA, and QBM.

5.1 System Design

The overall implementation followed a hybrid quantum-classical approach. The system architecture consisted of:

- Quantum circuits tailored for portfolio optimization (VQE), combinatorial problems (QAOA for MaxCut), and probabilistic modeling (Quantum Boltzmann Machines).
- Classical pre- and post-processing using NumPy and Pandas, integrated with quantum modules for data handling and optimization.
- Execution support for both simulators (default.qubit in PennyLane, Estimator in Qiskit) and potential quantum hardware backends.

5.2 Coding and Simulation

1. **VQE for Portfolio Optimization (VS Code):** Implemented using Qiskit in a local Python environment. Variational circuits were designed to minimize a custom Hamiltonian representing financial risk and returns. Execution was done using Estimator-based VQE with SPSA optimizer.
2. **QAOA for MaxCut (Google Colab):** Implemented using PennyLane. Graphs were encoded into cost Hamiltonians, and QAOA circuits alternated between cost and mixer unitaries. Parameters were optimized using classical gradient descent to solve MaxCut.
3. **Quantum Boltzmann Machine (QBM):** Implemented for generative modeling tasks using PennyLane. The QBM was trained using contrastive divergence and evaluated for its ability to approximate classical distributions.
4. **Installation and Setup:** All required installations for each framework (Qiskit, PennyLane, NumPy, etc.) are documented within their respective code files for reproducibility.

5.3 Execution Process

- VQE simulations were executed locally using Qiskit Estimator and custom ansatz circuits. Portfolio data (expected returns and covariance matrix) was fed into the quantum backend, and optimized weights were returned.

- QAOA and QBM implementations were executed using PennyLane’s `default.qubit` simulator in Google Colab, allowing parameter tuning and result visualization.
- All circuits were tested first on simulators to verify correctness before considering real quantum hardware execution.

5.4 Optimization Steps

To improve performance and feasibility:

- **SPSA (Simultaneous Perturbation Stochastic Approximation)** was used in VQE for efficient parameter optimization with minimal gradient evaluations.
- **Circuit depth reduction** was applied to ensure compatibility with current NISQ-era quantum devices and reduce simulation time.
- **Noise mitigation techniques** such as parameter reinitialization and shot averaging were explored during simulations.
- **Hyperparameter tuning** was performed on learning rates, ansatz structure, and optimization steps.

5.5 Tools and Frameworks

The following tools and frameworks were utilized:

- **Qiskit:** Used for implementing the VQE-based financial optimizer.
- **PennyLane:** Used for QAOA and QBM simulations on quantum circuits.
- **NumPy & Pandas:** Handled all classical data preprocessing and formatting for quantum input.
- **Google Colab:** Used for remote development and testing of PennyLane circuits.
- **VS Code:** Used for building and debugging the full FastAPI backend with VQE implementation.

5.6 Challenges and Mitigation

- **Dimensional mismatch in VQE inputs:** Resolved by validating the consistency between expected returns, fundamentals, and covariance matrices.
- **Quantum decoherence in simulators:** Managed by reducing circuit depth and simulating with shot noise to approximate real-device behavior.
- **Integration of quantum-classical pipelines:** Solved using structured modular design where classical computations (e.g., fetching stock data) were handled separately from quantum circuit evaluations.
- **Graph encoding complexity for MaxCut:** Simplified by using small example graphs for visualization and testing in QAOA.
- **Gradient computation instability in QBM:** Stabilized using PennyLane’s built-in optimizers and adaptive learning rates.

6 Conclusion

This report documented the exploration and implementation of quantum computing techniques using PennyLane, with a focus on their application in quantum finance and computational problem-solving. A diverse set of quantum algorithms was studied, implemented, and tested, including Variational Quantum Eigensolver (VQE), Quantum Approximate Optimization Algorithm (QAOA), Quantum Boltzmann Machine (QBM), Deutsch-Jozsa, and Shor’s algorithm.

The VQE algorithm was used to solve a realistic portfolio optimization problem, combining quantum circuit design with classical post-processing. All experiments and development were performed in both Google Colab and VS Code environments using PennyLane’s `default.qubit` simulator, allowing efficient prototyping of quantum-classical hybrid systems. The implementation included integration with NumPy for data processing and classical optimization.

QAOA was applied to the MaxCut problem using PennyLane, showcasing the algorithm’s effectiveness in solving combinatorial graph problems through parameterized quantum circuits. The implementation involved constructing cost and mixer Hamiltonians, optimizing variational parameters (γ , β), and sampling to extract high-probability solutions.

Quantum Boltzmann Machines (QBM) were explored as quantum generative models, enabling a new way of learning probability distributions from data using quantum energy functions. Although still in early stages of experimentation, the initial results indicate promise for quantum-enhanced machine learning.

In addition, the foundational quantum algorithms—Deutsch-Jozsa and Shor’s—were implemented to gain a deeper understanding of the fundamental capabilities of quantum computation. These served as benchmark experiments that verified the quantum simulator setup and illustrated the principles of quantum parallelism and quantum factorization.

All package installations and setup instructions were kept modular and organized within respective project files to ensure smooth reproducibility and debugging.

This work demonstrates a hands-on application of several quantum algorithms in the context of finance, optimization, and learning. It also highlights the capabilities of PennyLane in enabling hybrid quantum-classical workflows. Future work could include deploying these models on real quantum hardware and scaling them to larger, more complex financial datasets as quantum devices mature.

7 Bibliography

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